

Broadband transmit/receive circuit for NMR

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Measured energy in Japan
David von Seggern
(vosegg@seismo.unr.edu) University of Nevada
July 2012, page 10
DIGITAL OBJECT IDENTIFIER
<http://dx.doi.org/10.1063/PT.3.1619>
The article by Thorne Lay and Hiroo Kanamori is an excellent review of the 1994 Chilean earthquake. The authors state that the seismic energy released was approximately five times as much energy as that of a 100-megaton atmospheric nuclear device. I believe the authors used the relation for seismic energy release rather than total strain energy release. The seismic energy underestimates the total strain energy release by a variable that depends on the fault plane. Accounting for total strain energy release would increase the earthquake energy number by orders of magnitude. Despite the catastrophic damage potential of nuclear bombs, the forces of nature occasionally unleash much larger energy releases. Although the nuclear bombs are under our control, earthquakes, volcanic eruptions, and extreme weather events are not. However, by judicious preparation and avoidance measures, humans can significantly diminish the damage of natural events.

Comment on this article
By the act of hitting a ball with a bat, one calculates the force energy to deliver the ball to its new location, but one must also take into account that the ball extended its energy release to that which became struck by the ball as its momentum ceased and passed energy to the struck item. Therefore the parameters of the damage extend into the future when the received energy to that pushed upon, later becomes released in a new event. Perhaps calculations of one added that in, while another's calculations did not. E.Mc.
Written by Edgar McCarroll, 14 July 2012 19:59

An rf switch applies either the oscillator output or another external rf input (if desired) to the main rf gate. The gate is open as long as +5 V is present at the "gate in." A 5 V Zener diode clamps the input at +5 V and protects the gate. The gated rf output is applied to an adjustable rf attenuator set to -10.7 dB. This attenuator is not front-panel adjustable and limits the instantaneous power applied to the modulator to 80 W. Three additional rf attenuators mounted on the front panel permit adjustment of the rf power delivered to the modulator. When the net attenuation of these three rf

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- ² E. D. Jones, M. A. Palmer, and F. R. Franklin, Opt. Quantum Electron. **8**, 231 (1976).

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This note describes a circuit being used in a 5–60 MHz pulsed FT NMR system to deliver the transmitter pulse efficiently to the sample probe and then with good signal-to-noise ratio to couple the resulting NMR signal from the probe into the preamplifier.

To understand the principles of the circuit, first consider the conditions during the Receive time, when the transmitter pulse is not present. The 2.7-k Ω resistor delivers a 2.5 mA bias current to each PIN diode. The PIN diodes, therefore, become equivalent to resistors of approximately 10 Ω each,² so that they effectively ground that end of the broadband transmission line transformer.³ The 1N914 diodes are all open circuit and may be ignored. The transformer, therefore, behaves as an autotransformer and delivers the probe signal to the preamplifier via another transmission line transformer. If the probe's input impedance is 50 Ω , then the preamplifier will "see" a source resistance of 50 Ω . The preamplifier's input transistor is a Nippon Electric Company NC921 biased to give a low noise figure (maximum of 1.5 dB at 60 MHz) when driven

by any source resistance of from 50 to 150 Ω . The PIN diodes themselves do not contribute significant noise, as judged by experimentally short-circuiting them and noting no change in output noise level.

Next consider the conditions during the Transmit time. The transmitter voltage levels make all of the 1N914's conduct during part of the rf cycle. A rectifier circuit is made up of the 100-pF capacitor coupling to the 1N914

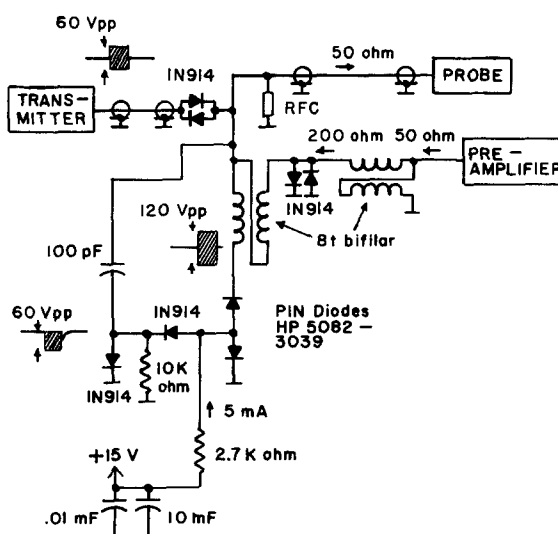


FIG. 1. Broadband Transmit/Receive circuit. During the Transmit time the PIN diodes are biased *off* and the rf power goes to the probe. During the Receive time, the PIN diodes short-circuit and the NMR signal flows from the probe to the preamplifier.

diode in parallel with 10 k Ω . This biases both PIN diodes off so that after a recovery time of about 0.2 μ s, they become high impedances. Negligible current will flow in either the primary or, as a result, in the secondary windings of the transmission line transformer. Most of the transmitter power, therefore, flows to the probe, to which it is impedance matched. The crossed diodes near to the input of the preamplifier are for its protection. The crossed diodes following the transmitter are for isolation during the Receive time. A moderate length of 50 Ω -cable is permissible between this circuit and the probe.

The transmission line transformers have a Z_0 of 100 Ω . They are 8 turns of a twisted pair wound on Stackpole 57-9322 ferrite toroid. The wire gauge is #28. The insulation diameter is 0.5 mm. A radio-frequency choke

(RFC) is shown which provides a dc path to ground for biasing of the upper PIN diode. In practice a filter may be present for other purposes, but which also provides the required dc path to ground.

The 10-W transmitter's power loss in going through the T/R circuit is about 1 dB. It would be even less at higher power levels. The noise figure measured at spots between 5 and 60 MHz is at most 3 dB. We expect that the frequency range could go farther, but did not have the amplifiers to test it.

This work was supported by NIH Grant GM 12446.

¹ T. C. Farrar and E. D. Becker, *Pulse and Fourier Transform NMR* (Academic, New York and London, 1971), pp. 37, 38.

² Unitrode Application Note MW-70-1.

³ C. L. Ruthroff, Proc. IRE 47, 1337 (1959).

Manual ellipsometry made semiautomatic

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A simple, inexpensive mechanical device is described that was designed to provide more effective data acquisition and to fit onto an already existing polarizer wheel drive mechanism of a commercially available, manual ellipsometer. A gas/metal adsorption experiment is used to illustrate typical results. Use procedures are detailed.

We have found that the data-taking process in our ellipsometric gas/metal adsorption studies¹ using a commercially available, manual ellipsometer² is made more effective by outfitting the polarizer (P) and analyzer (A) great circles with a simple and inexpensive mechanical device that permits relatively rapid measurement of phototube currents corresponding to P and A angles at light extinction using the off-null, equal-intensity technique.^{3,4} The device is shown schematically in Fig. 1. A lever (a) and gear (b) unit was used to drive each great circle reproducibly to either of two previously assigned settings. Lever-stop (c) settings were achieved by the pull of gravity or a small magnet (d). Set screws (e) were used to facilitate the necessary adjustments within a wide range of conditions. The device could be disengaged from and reengaged to a great circle and to previous P and A settings quickly without readjustment of the lever-stop settings.

Typically, the phototube signal was amplified using a picoammeter with current suppression and damping, and was recorded using a strip chart recorder. The lever-stops of each polarizer were set at the polarizer's minimum (value at extinction) and at a few degrees away from minimum⁴ so that during the course of an experiment one polarizer was at minimum while the other was off minimum; lever positions (a) were changed simul-

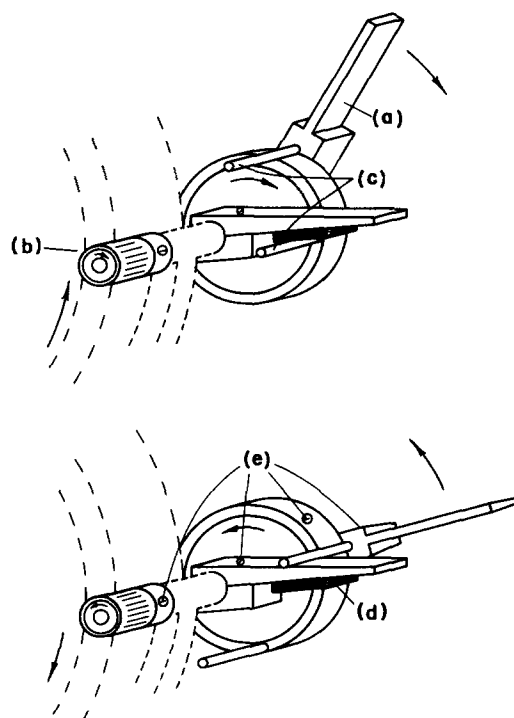


FIG. 1. Schematic diagram showing the device adapted to the movement mechanism of polarizer elements of a commercially available, manual ellipsometer.² Dotted lines indicate pre-existing parts and solid lines indicate new adaptations. See text for other details.